Summary in Graph

Exam Summary (GO Classes GATE CS/DA | DBMS | Relational Algebra | Weekly Quiz 11)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	10 (5 + 5)
Total Marks:	15 (5 + 10)
Exam Duration:	30 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained

Q1 Q2 Q3 Q4 Q5 Q6 Q7 Q8 Q9 Q10

* Average Stats based on first-attempt stats of Top 10% (44) out of 441 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Choice Type Award: 1 Penalty: 0.33 Databases

Consider the following schemas:

Branch=(Branch-name, Assets, Branch-city)

Customer = (Customer-name, Bank name, Customer-city)

Borrow = (Branch-name, Loan number, Customer account-number)

Deposit = (Branch-name, Account-number, Customer-name, Balance)

Using relational Algebra, the Query that finds customers who have balance more than 10,000 is ____

A. $\pi_{customer-name}(\sigma_{balance>10000}(Deposit))$
B. $\sigma_{customer-name}(\sigma_{balance>10000}(Deposit))$
C. $\pi_{customer-name}(\sigma_{balance>10000}(Borrow))$
D. $\sigma_{customer-name}(\pi_{balance>10000}(Borrow))$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 02sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Databases

With respect to relational algebra, which of the following operations are included from mathematical set theory?

i. Join
ii. Intersection
iii. Cartisian product
iv. Project

A. i and iv
B. ii and iii
C. iii and iv
D. ii and iv

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Databases

Given the relations

- employee (name, salary, dept-no), and
- department (dept-no, dept-name,address),

Which of the following queries cannot be expressed using the basic relational algebra operations $(\sigma, \pi, \times, \bowtie, \cup, \cap, -)$?

A. Department address of every employee
B. Employees whose name is the same as their department name
C. The sum of all employees' salaries
D. All employees of a given department

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Databases

Which operation is used to extract specified columns from a table?

A. Project
B. Join
C. Extract
D. Substitute

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Databases

The join operation can be defined as

A. a cartesian product of two relations followed by a selection
B. a cartesian product of two relations
C. a union of two relations followed by cartesian product of the two relations
D. a union of two relations

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Consider the relations $R(A, B)$ and $S(B, C)$ and the following four relational algebra queries over R ans S

I. $\pi_{A,B}(R \bowtie S)$
II. $R \bowtie \pi_B(S)$
III. $R \cap (\pi_A(R) \times \pi_B(S))$
IV. $\Pi_{A.R.B}(R \times S)$ where $R.B$ refers to the column B in table R

One can determine that:

A. I, III and IV are the same query
B. II, III and IV are the same query
C. I, II and IV are the same query
D. I, II and III are the same query

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Let R and S be two relations with the following schema

$R(\underline{P, Q}, R1, R2, R3)$

$S(\underline{P, Q}, S1, S2)$

where $\{P, Q\}$ is the key for both schemas. Which of the following queries are equivalent?

I. $\Pi_P(R \bowtie S)$

II. $\Pi_P(R) \bowtie \Pi_P(S)$

III. $\Pi_P\left(\Pi_{P,Q}(R) \cap \Pi_{P,Q}(S)\right)$

IV. $\Pi_P\left(\Pi_{P,Q}(R) - \left(\Pi_{P,Q}(R) - \Pi_{P,Q}(S)\right)\right)$

A. Only I and II
B. Only I and III
C. Only I, II and III
D. Only I, III and IV

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Select Type | Award: 2 | Penalty: 0 | Databases

Consider the following relation schemas:
$R(A, B),\ S(B, C),\ T(C, D)$.
Check equivalences that are correct under set semantics.
Assume that all set operations ($\cup, -$) are applied only on union-compatible relations (i.e., relations having the same set of attributes).

A. $R \bowtie R \equiv R$
B. $R \cup (S - R) \equiv S \cup R$
(Assume R and S are union-compatible.)
C. $(R \cup S) - T \equiv R \cup (S - T)$
(Assume R, S, and T are union-compatible.)
D. $\pi_A(R \bowtie S) \equiv \pi_A(S \bowtie R)$

Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Suppose that two relations $R(A, B)$ and $S(A, B)$ have exactly the same schema. Consider the following equalities in relational algebra?

I. $R \cap S = R - (R - S)$
II. $R \cap S = S - (S - R)$
III. $R \cap S = R$ NATURAL-JOIN S
IV. $\mathrm{R} \cap \mathrm{S} = \mathrm{R} \times \mathrm{S}$

Which of the equalities hold in relational algebra?
A. I only
B. I and II only
C. I, II and III
D. I, II, III and IV

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Consider the following instances:

QB1

Q	Ans
1	x
2	y
3	y

QB2

Q	Ans
1	x
1	y
2	y

Which of the following relational operations will produce the following tuple only?

3	y	1	y

A. $(\text{QB1} \cup \text{QB2}) \times (\text{QB2} \cup \text{QB1})$
B. $(QB1 - QB2) \cap (QB2 - QB1)$
C. $(\text{QB1} - \text{QB2}) \times (\text{QB2} - \text{QB1})$
D. $(QB1 \cap QB2) - (QB2 \cap QB1)$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

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